

Delsarte bound on binary code sizes

Om Joglekar

IIT Bombay

Guided by: Prof. Sudhir Ghorpade

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- By a *binary code of length n* we mean a set C of binary sequences of length n equipped with the Hamming distance such that $|C| \neq 0, 2^n$.
- **AIM:** To find the maximum size of C such that the distance between two codes is at least d .

Graph Theory background

Consider a simple graph $G = (V, E)$ on vertex set V with edge set E .

- By an ***independent set*** we mean a subset $S \subseteq V$ of vertices such that there are no edges between any two vertices of S .
- Given any $A \subseteq V$ we can define the binary vector $\chi(A) \in \mathbb{F}_2^V$ of length $|V|$, which we refer to as the ***characteristic vector of A*** , to be

$$\chi(A)_v = \begin{cases} 1 & \text{if } v \in A \\ 0 & \text{otherwise} \end{cases}$$

- For an independent set S , define a $|V| \times |V|$ matrix M to be

$$M = \frac{1}{|S|} \chi(S) \chi(S)^t$$

Graph Theory background (contd.)

Proposition 1

The matrix M is a positive semi-definite matrix satisfying

- $M_{v,w} = 0$ if $\{v, w\} \in E$
- $\text{trace}(M) = 1$
- $\sum_{v,w} M_{v,w} = |S|$

Proof.

The PSD property is trivial. We can in fact write down every entry of M as $M_{v,w} = \frac{1}{|S|} \chi(S)_v \chi(S)_w$.

$$M_{v,w} = \begin{cases} \frac{1}{|S|} & \text{if } v, w \in S \\ 0 & \text{otherwise} \end{cases}$$



Proposition 2

If $\alpha(G)$ denotes the independence number of the graph G , that is, the size of the largest independent set, then

$$\alpha(G) \leq \max_M \sum_{v,w} M_{v,w}$$

where M varies over all $|V| \times |V|$ PSD matrices such that $\text{trace}(M) = 1$ and $M_{v,w} = 0$ whenever $\{v,w\} \in E$.

The proof is obvious due to proposition 1.

Definition 3

A *Semidefinite Program (SDP)* is an optimisation problem of the form

$$\begin{aligned} & \text{maximise} && \langle C, X \rangle \\ & \text{subject to} && \langle A_i, X \rangle = b_i, \quad i = 1, \dots, m, \\ & && X \succeq 0, \end{aligned}$$

where $C, A_1, \dots, A_m \in \mathbb{S}^n$ are given symmetric matrices, $b \in \mathbb{R}^m$ is a given vector, and $X \in \mathbb{S}^n$ is the symmetric matrix variable constrained to be positive semidefinite ($X \succeq 0$).

Here $\mathbb{S}^n$ denotes the set of all $n \times n$ symmetric matrices over $\mathbb{R}$.

Semi-Definite Programming (contd.)

The bound we obtained is called the Lovász number introduced by Lovász to get bounds on the Shannon capacity of a graph [2].

More precisely,

$$\vartheta(G) = \max \left\{ \sum_{v,w} M_{v,w} \mid M \succeq 0, \operatorname{tr}(M) = 1, \right. \\ \left. M_{v,w} = 0 \ \forall \{v,w\} \in E \right\}$$

This is clearly an SDP (semi-definite program). All SDPs are solvable in polynomial time up to any desired error $\varepsilon > 0$. There are many algorithms to do this with the ellipsoid method being the most common one (also commonly used for linear programming).

Connecting graphs to codes

We construct the following graph G_d . Choose the set of vertices $V = \mathcal{B}(n)$, that is, the set of all 2^n binary codewords of length n . Let $x, y \in \mathcal{B}(n)$ be joined with an edge if and only if $d(x, y) \leq d - 1$.

Clearly any independent set in this graph will contain vertices with distance between any two being at least d . The aim has now shifted to finding the size of the largest independent set in this graph.

We can apply the bound obtained earlier which is computed by solving an SDP. However, the complexity of this bound in our case is not polynomial time because our graph G_d has 2^n vertices. Consequently, we have a polynomial in 2^n .

A ray of hope: We can however exploit the symmetry and in fact obtain a linear programming model.

Symmetry of the graph

Let us define H_n to be the set of all distance (Hamming) preserving bijections on $\mathcal{B}(n)$. This is just the group of isometries of the graph.

Proposition 4

The size of H_n is given by $|H_n| = 2^n n!$

Proof.

Every such distance preserving preserving bijection can be shown to be a permutation followed by flipping the parity of each entry individually. Any element of the group can be uniquely determined using the image of the origin $O = (0, \dots, 0)$ and each of the n vertices $e_i = (0, \dots, 0, 1, 0, \dots, 0)$ under this element. □

As a remark, H_n is in fact the automorphism group of the hypercube graph H_n on 2^n vertices. This proposition is not used directly but we care about the characterisation of maps in H_n .

Symmetry of the graph (contd.)

Given a binary code C of length n , we may define the *characteristic vector of C* to be the vector $\chi(C) \in \mathbb{F}_2^{2^n}$ given by

$$\chi(C)_x = \begin{cases} 1 & \text{if } x \in C \\ 0 & \text{otherwise} \end{cases}$$

We choose the dictionary ordering for the codewords:

$$(0, 0, \dots, 0) < (0, 0, \dots, 0, 1) < \dots < (1, 1, \dots, 1)$$

Lemma 5

For a binary code C of length n and any $\tau \in H_n$,

$$\chi(\tau(C)) = P_\tau \chi(C)$$

where P_τ is the $2^n \times 2^n$ permutation matrix corresponding to τ .

The lemma follows by comparing entries of the vectors on both sides.

Symmetry of the graph (contd.)

We define a matrix which is central to the rest of the theory.

Definition 6

Given a code $\emptyset \subsetneq C \subsetneq \mathcal{B}(n)$ define

$$M_C = \frac{1}{|C|} \sum_{\tau \in H_n} \chi(\tau(C)) \chi(\tau(C))^t$$

Although not relevant here, we can clearly see that this matrix is PSD.

We now recall the concept of the commutant. If V is an A -module, the set of all A -module homomorphisms from V to V is denoted $\text{Hom}_A(V, V)$. It is essentially the set of all linear maps $V \rightarrow V$ which commute with the action of A on V .

Symmetry of the graph (contd.)

Setting $A = \mathbb{C}[H_n]$ and $V = \mathbb{C}[\mathcal{B}(n)]$ we have

$$\text{Hom}_A(V, V) = \{M \in M_{2^n}(\mathbb{C}) \mid MP_\tau = P_\tau M \ \forall \tau \in H_n\}$$

Proposition 7

The matrix $M_C \in \text{Hom}_{\mathbb{C}[H_n]}(\mathbb{C}[\mathcal{B}(n)], \mathbb{C}[\mathcal{B}(n)])$.

Proof.

Using lemma 5 and $P_\tau^t = P_\tau^{-1}$ we get for any $\sigma \in H_n$

$$\begin{aligned} P_\sigma M_C P_\sigma^t &= \frac{1}{|C|} \sum_{\tau \in H_n} P_\sigma P_\tau \chi(C) \chi(C)^t (P_\sigma P_\tau)^t \\ &= \frac{1}{|C|} \sum_{\tau \in H_n} \chi((\sigma \circ \tau)(C)) \chi((\sigma \circ \tau)(C))^t = M_C \end{aligned}$$

□

The commutant and its orbital basis

Let a group G act on a set X . Let the permutation matrix of the bijection given by $g \in G$ of size $|X| \times |X|$ be denoted by $\rho(g)$.

We define $\text{Hom}_G(S, T) = \{A \in M_{|T| \times |S|}(\mathbb{C}) \mid A\rho(g) = \tau(g)A\}$. This is a vector space over $\mathbb{C}$. If $S = T$, then it becomes a $\mathbb{C}$ -algebra with matrix multiplication.

Note the commutant we had earlier may simply be written as $\text{Hom}_{H_n}(\mathcal{B}(n), \mathcal{B}(n))$.

Theorem 8

Let $M \in M_{|T| \times |S|}(\mathbb{C})$. Then $M \in \text{Hom}_G(S, T)$ if and only if

$$M(t, s) = M(g * t, g * s) \quad \forall t \in T, s \in S, g \in G$$

The above theorem is easily proved by comparing entries on both sides of the equality $M\rho(g) = \tau(g)M$.

The commutant and its orbital basis (contd.)

Definition 9

Let G act on sets T and S and hence on $T \times S$ by extending the action. Let $\theta_1, \dots, \theta_m$ be the orbits of this action on $T \times S$. Define the matrices $M_j \in M_{|T| \times |S|}(\mathbb{C})$ for $j = 1, 2, \dots, m$ to be

$$M_j(t, s) = \begin{cases} 1 & \text{if } (t, s) \in \theta_j \\ 0 & \text{otherwise} \end{cases}$$

The set of these m matrices is called the ***orbital basis*** of $\text{Hom}_G(S, T)$.

The fact that the orbital basis forms a basis is easy to prove. This also shows that the dimension of the commutant is the number of orbits of the action on the product of the sets.

Lemma 10

Let a group G act on a set S and let $M_1, \dots, M_m$ be the orbital basis of $\mathcal{A} = \text{Hom}_G(S, S)$. If each M_j is symmetric, then the $\mathbb{C}$ -algebra $\mathcal{A}$ is commutative.

The proof is trivial since every matrix will be a linear combination of the symmetric matrices M_i and one can show that the basis itself commutes, that is, $M_i M_j = M_j M_i$. Alternatively,

$$MN = (MN)^t = N^t M^t = NM$$

The commutant and its orbital basis (contd.)

Theorem 11

$\text{Hom}_{H_n}(\mathcal{B}(n), \mathcal{B}(n))$ is a commutative algebra of dimension $n + 1$.

The proof follows from Gelfand's lemma 10 and studying the orbits of the action of H_n on $\mathcal{B}(n) \times \mathcal{B}(n)$.

In particular one observes that (α, β) and (α', β') lie in the same orbit if and only if $d(\alpha, \beta) = d(\alpha', \beta')$. This distance can take values from 0 to n giving the required dimension and $d(\alpha, \beta) = d(\beta, \alpha)$ gives the commutativity.

The commutant and its orbital basis (contd.)

Coming back to our setting we have the orbital basis $\{A_0, A_2, \dots, A_n\}$ given by

$$A_i(x, y) = \begin{cases} 1 & \text{if } d(x, y) = i \\ 0 & \text{otherwise} \end{cases}$$

Define $\lambda_i = |\{(x, y) \in C \times C \mid d(x, y) = i\}|$. Normalise it further to

$$\alpha_i = \frac{\lambda_i}{|C| \binom{n}{i}}$$

so that $\alpha_0 = 1$.

Now observe that

$$\sum_{i=0}^n \alpha_i \binom{n}{i} = \frac{1}{|C|} \sum_{i=0}^n \lambda_i = |C|$$

The commutant and its orbital basis (contd.)

Proposition 12

$$M_C = n! (\alpha_0 A_0 + \cdots + \alpha_n A_n)$$

Proof.

We simply count the number of 1's on both sides of the equation. On the right hand side we may add up the counts because all the A_i 's have disjoint supports.

The number of 1's in A_i is simply $2^n \binom{n}{i}$. This is obtained by counting the 1's column-wise: for each one of the 2^n columns corresponding to a codeword, we may flip any i coordinates to obtain a new codeword with distance i from it.

The number of 1's in $\chi(C)\chi(C)^t$ in the positions fixed by the 1's of A_i is λ_i since $(\chi(C)\chi(C)^t)(v, w) = 1_{v \in C} 1_{w \in C}$ and we restrict to only those (v, w) satisfying $d(v, w) = i$.

The result now follows by the definition of α_i . □

Summarising into the main result

Theorem 13

If C is a binary code of length n with minimum distance d , we have

$$|C| \leq \max \sum_{i=0}^n \binom{n}{i} x_i$$

subject to

- ❶ $x_i \geq 0 \quad \forall i = 0, \dots, n$
- ❷ $x_0 = 1$ and $x_1 = x_2 = \dots = x_{d-1} = 0$
- ❸ $x_0 A_0 + \dots + x_n A_n$ being PSD

This theorem simply uses the observation made just after defining the α_i which are now replaced by the optimising variable x_i .

This is still not enough for the PSD condition cannot be solved for in polynomial time because the size of the matrices is still 2^n .

Eigenvalues

However, if we were to know the eigenvalues of $\sum x_i A_i$ then we would be done since a matrix is PSD if and only if all its eigenvalues are non-negative and this condition is clearly linear in the x_i .

Let the standard basis vectors of $\mathbb{C}[\mathcal{B}(n)]$ be $\{e_X \mid X \subseteq [n]\}$ with $2^{[n]}$ being ordered dictionary-wise. This is indeed a $\mathbb{C}$ -basis of $\mathbb{C}[\mathcal{B}(n)]$. Now we can decompose this vector space as

$$\mathbb{C}[\mathcal{B}(n)] \cong \mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2$$

The isomorphism (vector space isomorphism over $\mathbb{C}$) is given by $e_X \mapsto a_1 \otimes \cdots \otimes a_n$ where

$$a_i = \begin{cases} \hat{1} = (0, 1)^t \in \mathbb{C}^2 & \text{if } i \in X \\ \hat{0} = (1, 0)^t \in \mathbb{C}^2 & \text{otherwise} \end{cases}$$

Eigenvalues (contd.)

Lemma 14

A_i acts on $a_1 \otimes \cdots \otimes a_n$ in the following way. We choose any i coordinates and flip them to obtain a new tensor product. Finally add all such $\binom{n}{i}$ tensor products to get the result.

Proof.

Let us start with $e_X \in \mathbb{C}[\mathcal{B}(n)]$. We know that the action of A_i is given by $A_i e_X = \sum_{Y:d(X,Y)=i} e_Y$. Now e_X corresponds to some $a_1 \otimes \cdots \otimes a_n$. Given a Y at a distance of i from X , if Y corresponds to $b_1 \otimes \cdots \otimes b_n$, then it is clear that some i of the a_i have been flipped to obtain the b_i . The result follows. $\square$

Eigenvalues (contd.)

For those who seek formalism, A_i can be realised as the operator:

$$\sum_{\substack{S \subseteq [n] \\ |S|=i}} X_S$$

where $X_S = M_1 \otimes \cdots \otimes M_n$ with

$$M_k = \begin{cases} X & \text{if } k \in S \\ I & \text{otherwise} \end{cases}$$

where

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} ; \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Eigenvalues (contd.)

Fix any $j \in \{0, 1, \dots, n\}$. Define a space

$$W_j = \text{span}(\{u_1 \otimes \cdots \otimes u_n \mid j \text{ of the } u_i \text{ are } v \text{ and the others are } u\})$$

We shall show that this is an eigenspace and find the eigenvectors.

Eigenvalues (contd.)

Proposition 15

Each W_j is an eigenspace of A_i of dimension $\binom{n}{j}$.

Proof.

The dimension is trivial to find. Let us take a tensor $w = u_1 \otimes \cdots \otimes u_n \in W_j$ and act A_i on it. From lemma 14 we know the action of A_i . Now the key point is that u ‘flips’ to u and v ‘flips’ to $-v$. Thus for every i coordinates we choose to flip, the vector remains the same and we can pull out the sign $(-1)^{\#v \text{ flipped}}$.

Thus if J denotes the positions of w where we have a v , we have

$$A_i w = \left(\sum_{S \subseteq [n], |S|=i} (-1)^{|S \cap J|} \right) w$$

This shows that W_j is indeed an eigenspace. □

Eigenvalues (contd.)

Proposition 16

The W_j are all the eigenspaces of A_i and the eigenvalue is

$$\lambda_j = \sum_{k=0}^i (-1)^k \binom{j}{k} \binom{n-j}{i-k}$$

Proof.

We can either see the first fact through the dimensions of each W_j or we can simply use Theorem 11. In any case, the eigenvalue is

$$\begin{aligned} \sum_{S \subseteq [n], |S|=i} (-1)^{|S \cap J|} &= \sum_{k=0}^i (-1)^k |\{S \subseteq [n] : |S| = i, |S \cap J| = k\}| \\ &= \sum_{k=0}^i (-1)^k \binom{j}{k} \binom{n-j}{i-k} \end{aligned}$$



Conclusion

Note that the A_i belong to a symmetric commutative algebra of dimension $n + 1$ and are thus simultaneously diagonalisable. Thus they share the same eigenspace decomposition.

Suppose the $n + 1$ distinct eigenvalues of A_i are $\lambda_0^{(i)}, \lambda_1^{(i)}, \dots, \lambda_n^{(i)}$. Then the eigenvalues of $\sum_{t=0}^n x_t A_t$ are

$$\mu_j = \sum_{t=0}^n x_t \lambda_j^{(t)} \quad \forall j = 0, 1, \dots, n$$

each of multiplicity $\binom{n}{j}$ corresponding to the eigenspace W_j . This finally reduces our task to a linear programming problem which can now be solved in polynomial time.

Conclusion (contd.)

Finally, we have the following theorem which is our conclusion.

Theorem 17

The size of a binary code C of length n with minimum distance d is bounded above by

$$\max \sum_{i=0}^n \binom{n}{i} x_i$$

where the maximum is taken over all $x_0, x_1, \dots, x_n$ such that

- ❶ $x_i \geq 0 \forall i = 0, \dots, n$
- ❷ $x_0 = 1$ and $x_1 = \dots = x_{d-1} = 0$
- ❸ $\sum_{i=0}^n \sum_{k=0}^i (-1)^k \binom{j}{k} \binom{n-j}{i-k} x_i \geq 0 \forall j = 0, 1, \dots, n$

- [1] P. Delsarte. “An algebraic approach to the association schemes of coding theory”. In: *Philips Research Reports Supplements* 1973.Suppl. 10 (1973).
- [2] L. Lovasz. “On the Shannon capacity of a graph”. In: *IEEE Transactions on Information Theory* 25.1 (1979), pp. 1–7. DOI: [10.1109/TIT.1979.1055985](https://doi.org/10.1109/TIT.1979.1055985).

Thank You!

joglekar.om@iitb.ac.in